

Course title: "Structure Machine1"	Calculator: Not allowed
Level: Year 1 (LMD)	Phone: Not allowed
Final Exam 2024-2025 (01 h 30 min)	
Full Name:	Student Number: Group:

Exercise 1: Comprehension questions (5 pts)

Q1- Give in decimal the range [min, max] in which a signed integer can be written on 8 bits in: (1,5 pts)

SAV	One's complement	Two's complement
[,]	[,]	[,]

Q2-Is it possible to use an XOR logic gate to generate an inverter? (1 pt)

Yes	No
Give the equivalent circuit:	Justify:

Q3-Why is gray code used to number rows and columns in the Karnaugh table? (1,5 pts)

Q4-What is the purpose of SEM normalization of floating point numbers? (1 pt)

Exercise 2(5 points)1) Let the integers in decimal, $A = -3$ and $B = -5$. Give, in binary on 4 bits, $A+B$ in: (3pts)

If a result is not correct, suggest a solution.

Suggestion:	Suggestion:	Suggestion:
SAV	One's complement	Two's complement

2) $(110,11 / 11,1)_2 = (?)_2$, with 5-bit precision. (1 pt)

3) Give the natural binary code of the Gray code: 1010011101 (1 pt)

Exercise 3

A-Suppose floating point binary numbers are encoded in the [Sign, Exponent, Mantissa] format as follows: 1-bit sign (S), 8-bit exponent (E), 8-bit mantissa (M), and the bias (B) is 127.

1) Let $x_1 = 11,625$ in base 10. Convert x_1 to binary in the [S,E,M] format described above.(0,25+0,25+0,5) pt

S	E	M

2) Let $x_2 = -3,5$ in base 10. Convert x_2 to binary in the [S,E,M] format described above. (0,25+0,25+0,5) pt

S	E	M

3) Calculate $x_3 = x_1 + x_2$ in [S,E,M] (0,25+0,5+0,75) pts

S	E	M

B-ASCII to text conversion (1,5 pts)

Convert the following ASCII code to its equivalent text.

73 32 100 111 110 39 116 32 99 104 101 97 116 46

Recall: In the decimal ASCII code table: the alphabetic characters are coded starting from A = 65. The decimal numbers are coded starting from zero = 48. Space = 32, period = 46, apostrophe = 39, and comma = 44.

Exercise 4 (5 pts)

Let the function $F(A,B,C,D)$ be defined by the following minterms:

$$F(A,B,C,D) = \sum (0, 1, 4, 6, 8, 9, 10, 12, 14, 15)$$

1. Give the truth table of the function F. (0,5 pt)

Truth Table (Questions 1 and 2)	Karnaugh map (Question 3)
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A	B	C	D	F

CD AB				

2. Give the 1st and 2nd canonical forms of the function from the truth table. (0,5 + 0,5) pt

1 st form	F =
2 nd form	F =

3. Simplify F by Karnaugh's method and obtain the most simplified form. (0,5 + 0,5 + 0,5) pts	
1's grouping	0's grouping
F =	F =
The most simplified form is:	

4. Use NAND logic gates only to represent the most simplified form of F (0,5 pt + 0,5 pt)
F = Circuit:

5. Use NOR logic gates only to represent the most simplified form of F (0,5 pt + 0,5 pt)
F = Circuit: